Lunar Lions

Software Proposal for NASA SUITS 2026 Design Challenge

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I. Technical Section

A. Abstract

The Lunar Lions SUITS team plans to develop an augmented reality interface for the Magic Leap 2 headset and a display and control system for the rover operating in DUST to support Artemis astronauts during lunar EVAs. The interface uses a dynamically adjusting UI, audio and visual cues, and gesture recognition to create an intuitive simple user experience. An integrated Artificial Intelligence Assistant (AIA) provides real-time, context-aware support, guiding astronauts through tasks such as navigation, LTV repair, and egress/ingress. It also facilitates interoperability between the Spacesuit the Pressurized Rover through standardized communication protocols, enabling real-time sharing of telemetry, hazard alerts, and annotations across mission assets.

Our design will be validated through staged human-in-the-loop tests, beginning with basic AR displays and progressing to full integration of dynamic telemetry and AI-guided operations. A user manual and guided tutorial will ensure intuitive operation, demonstrating how the AR interface and intelligent assistance can enhance astronaut performance, safety, and collaboration on the lunar surface.

B. Software and Hardware Design Description

Team Interoperability

To ensure seamless inter-team interoperability as outlined in our mission objectives, our design focuses on efficient data exchange and communication between the Magic Leap 2 spacesuit UI and the pressurized rover (PR) control system. The EV and PR team will closely work together to establish standardized protocols, data formats, and mechanisms that enable real-time sharing of critical information, coordinated actions, and mission support. By using a unified communication protocol based on JSON over WebSockets, we maintain full compatibility between the Magic Leap 2 and PR systems. For example, the PR will transmit hazard updates during EV navigation while the EV sends LTV status updates following repairs.

During the test week, NASA's local network will support this data exchange without additional hardware, as we continue collaborating with the PR team to define consistent telemetry data schemas. This configuration also enables the sharing of EV and PR location coordinates, spacesuit telemetry (video feed, oxygen levels, battery status, etc.), biomedical data (heart rate, body temp, etc.) from EV to PR, system telemetry (resource levels and navigation status), and environmental data from PR to EV.

Our design also integrates robust alerting, mapping, and AI assistance to strengthen coordination. Alerts from the PR system appear as peripheral notifications on the EV's HMD. On the PR interface, EV-generated alerts are prominently displayed, enabling the PR team to monitor EV safety and provide assistance as needed. Additionally, a synchronized 2D minimap allows both EV and PR teams to add, remove, and annotate locations in real-time through voice commands or gestures on the Magic Leap 2 and directly on the PR interface. Furthermore, the EV's camera feed will be efficiently streamed to the PR display using adaptive compression protocols, ensuring smooth video playback and allowing the PR team to effectively monitor EV activities. The Artificial Intelligence Assistant (AIA) supports this workflow by filtering

telemetry, prioritizing alerts, and guiding the EV through shared procedures such as navigation and LTV repair.

Pressurized Rover Navigation

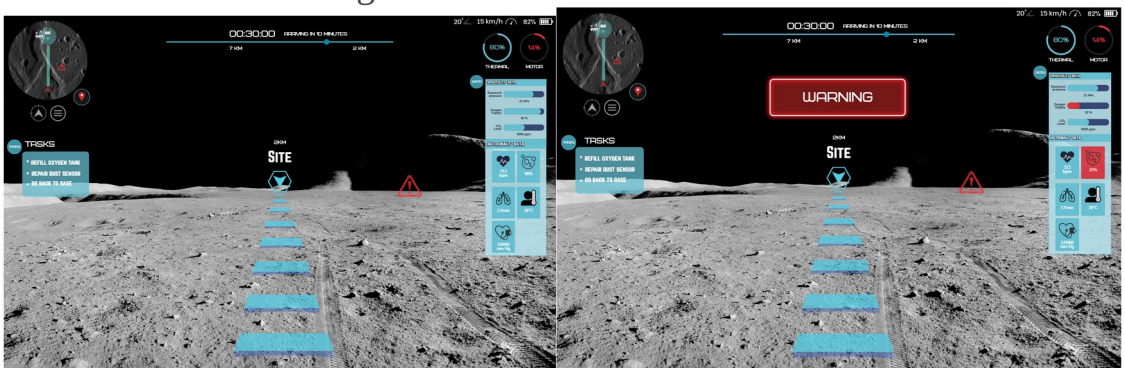


Fig 1. Rover UI during navigation and warning system

Navigation Algorithm

The Pressurized Rover (PR) begins with incomplete knowledge of its environment. Thus, we implement a D* Lite algorithm to iteratively gather information and construct an optimal path as it gains information about its surroundings. The environment is modeled as a weighted graph (nodes representing locations and edges representing travel costs) and initialized to be obstacle free. As new obstacles are detected, their costs are backpropagated through the graph, allowing the rover to replan in a computationally fast and memory-efficient manner [1]. The cost function g(n) accounts for spatial distance, resource usage, and slope, while a one-step lookahead function rhs(n) estimates the minimum cost to the goal through neighboring nodes. The algorithm compares g(n) and rhs(n) to identify inconsistencies and revise the optimal path. Our heuristic h(n) uses Euclidean distance between node n and the start node, assuming a flat surface, an admissible approach since the rover is omnidirectional. When hazards are detected, their corresponding nodes are assigned a large cost so the rover naturally avoids them.

Search and Rescue

The PR will first navigate toward the LTV's last known location using D* Lite. Once there, it will conduct an expanding square search within a radius defined by the LTV's consumable limits, calculated from its maximum speed and battery capacity. The PR will move in 950 m increments, below the 1000 m signal diameter, to ensure no beacon signals are missed. Upon detecting the LTV signal, the PR will perform a finer expanding square search within a 500 m radius using 10 m increments, leveraging LIDAR and beacon data to pinpoint the LTV. D* Lite will continue managing obstacle avoidance during this search. When within 50 m of the LTV, the interface will display a proximity indicator with GPS coordinates and emit a progressively faster beeping sound, guiding the rover toward the target.

Hazard Identification and Avoidance

Environmental data for D* Lite will be provided by computer vision systems processing live rover camera feeds to detect and classify hazards (Fig. 1), using data from the Telemetry Stream Server (TSS) server. Once a hazard is identified, its size and position are recorded, and the affected node is assigned a high traversal cost. The rover then recalculates the path.

Resource Tracking

Before travel, the PR will verify that its power and life-support resources are sufficient for the planned journey. If the projected cost exceeds available capacity, the system will abort

the route and notify the operator. D* Lite will then identify turnaround points within a defined maximum cost, ensuring the rover can safely return, if needed [2]. Hazards detected by either rover or crew cameras will automatically update the shared map for safer future path planning.

EV Navigation

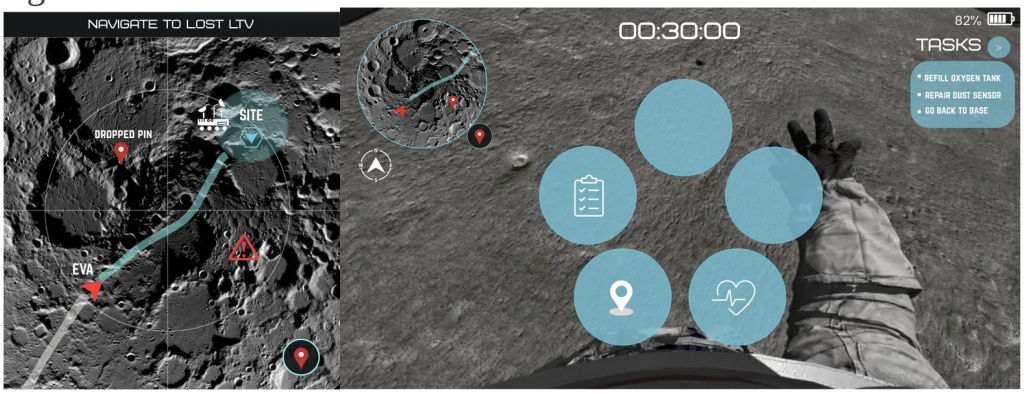


Fig 2. Spacesuit UI during EV navigation

Best-Path Navigation

We will use D* Lite to calculate the optimal path for the EV, with modifications to account for differences in mobility between the EV and PR and to incorporate human vitals. The algorithm will include maximum travel cost constraints based on biometric data and life-support limits, similar to what was done for PR.

Drop Pin Sharing

The PR and EV can share pins through the minimap feature. When a user drops a pin, their position from the TSS server will be mapped to a coordinate on the shared map. The pin first appears on the user's local UI, then syncs ok with the server, which updates the global pin set across both teams. The minimap will display two persistent pins marking the current positions of the PR and EV.

Resource Tracking

Similar to PR procedure, when an LTV is detected or a location pin is selected, the EV will be shown a best path calculated by the D* algorithm, provided vitals (e4.g., oxygen and battery levels) from TSS are sufficient to reach the destination.

Breadcrumb Path

A breadcrumb system will allow EVs to retrace their steps to the PR. The system places markers at fixed intervals, forming a continuous path anchored to the terrain. Overlapping markers within a short distance (e.g., 1–2 m) are merged to reduce clutter, and each marker includes a timestamp that fades in opacity or color over time to indicate recency. An overhead compass with waypoints will indicate heading and help EVs orient themselves.

Asset Tracking & Hazard Detection

Using telemetry data from the TSS, the AR minimap continuously displays the positions of lunar assets, crew members, and scientific points of interest, each marked with distinct icons and updated in real time. As crew members and rovers move, onboard sensors and vision systems detect impassable terrain and hazards, which are fed into the D* path-planning algorithm to dynamically recalculate the safest route. The minimap updates in parallel, displaying new hazards and the revised navigation path. In 3D mode, elevation data provides additional spatial awareness by showing the relative heights of assets and hazards.

AR Interface Overview

We plan to use the Magic Leap 2 headset for the EVA phase to provide an advanced AR interface. Using the Unity XR Interaction Toolkit for customizable elements, we will create a UI that adapts to mission phase and user interactions, whether biometrics, procedures, navigation, or more. Certain alerts will be auditory, such as warning tones when biometric data exceed safe limits. Navigation modes will support tactile interaction (virtual pointing, clicking, dragging), gaze control, shortcut gestures, and voice commands. This multimodal interaction allows EVs to use the most intuitive method at any moment.

To reduce cognitive load, we will employ a unified software platform design. For example, if the EV is navigating to a site, the UI will suppress unrelated interactions, keeping focus on current objectives. The interface also supports real-time communication between the EV and PR through shared waypoints, annotations, and status updates.

To address low-light conditions during lunar EVAs, EV crew members will use high-intensity, head-mounted flashlights for illumination. The AR interface will feature adaptive brightness and contrast control, with critical UI elements highlighted in luminescent colors. Together, these enhancements ensure situational awareness under minimal illumination.

UI Components

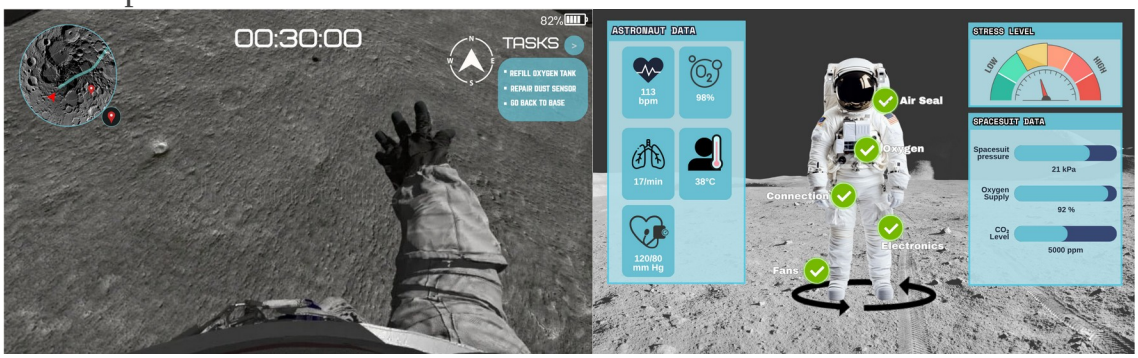


Fig 3. Spacesuit system and biological data interface

Constants

Constant components will be displayed around the upper edges of the EV's field of view to avoid obstructing vision near the feet and center. They will be hideable using voice commands, gestures, or the main menu. These components include current objectives, elapsed mission time, a 2D minimap, and a compass.



Fig 4. System warning

Discrepancies in health-related data will initially be reflected by warning tones. When life support system metrics approach dangerous levels, a notification will pop up in the center of the UI. To avoid obstructing the EV's view, the notification will then minimize to the side. All alert notifications will be compressed into a button that, when pressed, displays the current alerts in a list. Message notifications from the pressurized rover team will drop down from the top of the screen and can be expanded via voice commands or by pressing on them or dismissed by clicking the "x" close button at the top. The main menu can be summoned by holding out the

palm. It contains buttons for components such as the telemetry data, procedures, navigation, settings, and a menu of organizational templates for the aforementioned views. All menus can be displayed simultaneously and are easy to resize and move using default touch interaction mechanisms. Screens dynamically snap in place in relation to other screens and can be locked together and follow the EV around, further organizing the information displayed.



Fig 5. Main menu

In addition to the always-available hand menu, voice control is available for functions like opening/closing menus or templates, navigating back, clearing the screen, dropping pins, performing other gestures, and asking the AI assistant questions.

Gestures

Our design incorporates intuitive hand gestures to enhance user interaction with the AR interface. Leveraging the Magic Leap 2's gesture recognition, EVs can perform actions such as opening menus, selecting options, and switching between templates with simple hand movements. For example, holding out the palm summons the main menu, and specific gestures allow quick access to functions like dropping pins or toggling UI components. This control minimizes cognitive load and physical strain, enabling EVs to interact with the system efficiently while keeping their focus on mission tasks.

Egress and Ingress Assistance

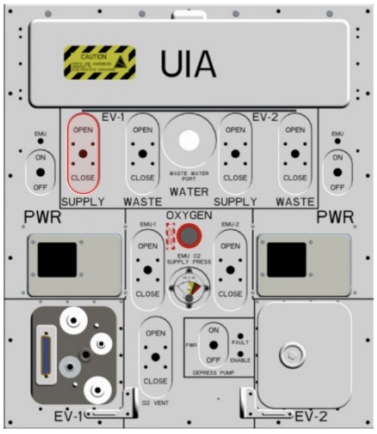


Fig 8. UIA with overlaid switch highlights.

For egress and ingress, we will use computer vision tools to recognize the contours of the UIA and overlay visual cues highlighting which buttons and switches to interact with at each step. The display updates as switches are flipped, while audio narration guides each procedure. A collapsible task list presents the UIA procedures step-by-step, expanding dynamically to minimize clutter and maintain focus. Each step corresponds to telemetry feedback from the TSS, with successful actions automatically confirmed through audio and visual checkboxes.

The EV can progress hands-free through voice commands to advance tasks, check system status, or expand sections. If clarification is needed, the EV can query the AI assistant for terminology explanations or troubleshooting. Incorrect, skipped, or out-of-order steps trigger immediate visual and audio warnings, and the AIA provides guidance for backtracking or restarting as needed. This system ensures procedural accuracy while minimizing manual input, enhancing both safety and efficiency.

LTV Repair

Architecture

Fig. 7 LTV architecture in Appendix C

Overview

Upon arrival at the malfunctioning Lunar Terrain Vehicle, the EV will engage with the AIA to conduct ERM, system diagnosis, restart, and final system checks and repair verifications. The high-level process for diagnosis, restart, and final checks is outlined in Concept of Operations (CONOPS). Once the AIA initializes system diagnosis, the headset will begin extracting snippets of video feed using on-headset softwares (MLCamera API or Spectator App). These video snippets will then get packaged and sent to a server, which then goes through a vision processing pipeline such as the one pictured in Fig. 7.

Vision Processing Model

Visual features will be extracted from the video snippets using Meta's DINOv3 model, a state-of-the-art (SOTA) Computer Vision model, on a backend server. We choose DINOv3 [9] because the model is accurate, efficient, and customizable. Every sub-1B parameter vision transformer tower achieved high performance on benchmarks pre-fine tuning at relatively low computational cost [9]. And, the model is easy to modify with extra convolutions/linear layers [13]. Taking advantage of these benefits, we will first split the video snippet into individual frames, attach a temporal head to relate the frames of the video to each other parametrically, and then run them through the model, extracting rich features distinguishing specific parts of the LTV (ex., Task board, wire locations, specific switches, etc). The visual features distinguish the specific parts, but don't give information on what the parts are and whether they're broken. This context can be obtained from NASA's training data, the TSS, the ERM data, AIA's thought process, and the EV's voice snippets. Thus, we plan to pass the visual features through two small classification heads [10], which convolute the visual tokens based on a pool of context [12], classify the features as specific parts, and establish if they're broken [11**].** Both heads are small linear classification layers, trained for the purpose of identifying broken regions. This overall process helps us bypass the limitations of data-availability; the larger DINOv3 backbone stays frozen to simply extract general visual features, while the smaller linear heads can be trained more efficiently and effectively on the less labelled data.

Data Cleansing

Dealing with data constraints: Information that is either obscured or completely not visible to the visual model will be supplemented by the TSS. The TSS will provide data as to which components are in disrepair in the damaged system, and this information when communicated to the visual model will allow it to confirm its IDs or fill in missing IDs. Finally, AIA information can be utilized to confirm the IDs of the visual model, either by its interpretation of the outputs, by EV confirmation of the diagnosis, or the EV directly, verbally stating what is in disrepair which can then be referenced against the visual model's IDs.

Model Output to AIA

The pipeline's final outputs are a series of absolute coordinates, and their respective damage levels (severe damage, non-negligible but non-threatening damage, fixed-and-working). This information gets reported back to the AIA in strict JSON format, from which the information will be displayed on the headset and the system diagnosis will conclude. A possible output is colored fixed-point spheres, placed on the LTV based on these coordinates. During physical repair tasks, on-headset computer vision libraries like Barracuda, OpenCV, and YOLOv8 can be used to box important objects, parts of the LTV, and more, reducing EV cognitive load. The instructions for physical repairs can also be used as more context for the system diagnosis. The final system check can be completed using the same pipeline, as after diagnosis, no anchors should light up as red.

Knowledge Graph

We will implement a knowledge graph such as the one pictured in Fig. 6 to organize and present mission-related data in a structured and intuitive manner. The knowledge graph connects various mission elements such as assets, crew members, procedures, geological info, and telemetry data. The user will be able to select nodes to reveal additional information and relationships. For example, selecting an LTV repair will allow the EV to locate the vehicle's location, run computer vision, and communicate with the AIA to receive procedures for repairing the LTV. This interconnected data structure enhances situational awareness and decision-making by providing context-rich information at a glance. Additionally, the underlying graph database will provide an organized data structure to train our AI assistant on the mission.

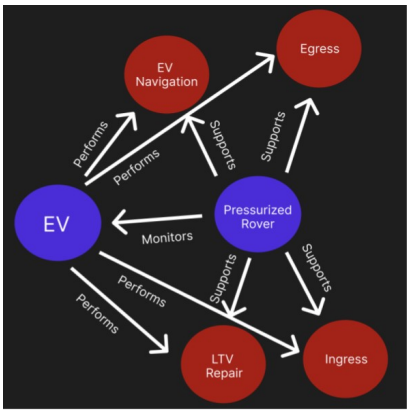


Fig 6. An example SUITS knowledge graph

C. Concept of Operations (CONOPS)

Egress

The mission begins with safe and efficient EV egress from the habitat. Inside the airlock, the crew member dons the HMD, spacesuit, and EVA tools before activating Egress Mode. Once engaged, the HMD transitions into an interactive interface guiding the EV through all egress procedures. The crew member will complete the set of UIA Egress procedures with the assistance of the UI indicators, audio cues, task list, and AI voice assistant. Once all steps are completed and verified by TSS data, the crew member can begin the EVA navigation.

EV Navigation

After completing egress, the HMD transitions to the navigation state. This mode prioritizes situational awareness with customizable widgets and adaptive brightness for lunar low-light conditions. The EV follows the AR interface to avoid obstacles and navigate to the LTV and points of interest. The EV interacts with its mini-map to view location, avoid hazards, and drop or view pins. Thermal infrared cameras generate a heat map of the surrounding terrain, while computer vision detects and classifies hazards such as rocks and craters.

The program autonomously updates routes in real-time and provides alerts for hazards, route deviations, or resource concerns. The EV verbally interacts with the AIA which leverages pathfinding to compute optimal routes. The interface displays multiple route options and dynamically draws paths using computer vision overlays that the crew member follows.

Pressurized Rover Navigation

The rover navigates to find the LTV and to reach areas of geologic interest, calculating the safest and most efficient route while updating paths as new hazards appear. It also estimates turnaround points based on resource availability to ensure safe return to the habitat. The rover's interface monitors EV vitals and system resources in real time. If vital signs or battery levels fall below safe thresholds, the system alerts the user to initiate return protocols.

LTV Repair

Upon reaching malfunctioning LTV, the crew member commands the AIA to begin ERM procedures and waits for audio and visual confirmation of completion. System diagnostics is then run by the AIA. Anchors are rendered in the UI, and AIA guides EV through visual inspection and repair tasks. AIA directs the restart of the rover navigation system and cross-checks post-restart telemetry with TSS data. If issues are detected, AIA alerts the EV to correct the mismatch. This process iterates until all warnings are cleared. After repairs, AIA instructs a final diagnostic to verify system stability, with the crew member confirming completion once all checks are done.

Ingress

After LTV repair, the crew member enables Ingress Mode, which displays the shortest and safest return route using both an AR breadcrumb trail and a mini-map. Upon reaching the habitat, an UIA Ingress Procedures checklist automatically activates. This collapsible task list mirrors the egress interface, guiding re-entry procedures with telemetry-based confirmations and audio/visual cues. As in egress, incorrect step notifications, UI highlights, and AI voice assistance ensure proper procedure adherence. Once all objectives are complete and systems nominal, the EV operates the airlock to re-enter safely.

Continuous Tasks

The EV can attach current tasks and suit data to the main display via the hand menu. This includes oxygen, CO_2, pressure, temperature, humidity, and mission timer data. The timer provides periodic updates to help EV track task durations and overall mission time. By minimizing on-screen icons and offering quick access to critical information, this menu supports efficient task management throughout the EVA. The AIA also continuously monitors life support and environmental telemetry to predict safe travel ranges and warn when limits approach. If biometric or environmental data move outside nominal ranges, the system issues alerts with recommended corrective actions.

D. Artificial Intelligence

Architecture

Fig. 9 System Architecture in Appendix C

Overview

The Artificial Intelligence Assistant (AIA) functions as both a personal aide, task assistant, and a critical safety monitor. It features a voice-activated chatbot that leverages real-time data, such as vitals, location, and system status, to provide context-aware responses. A local LLM will be hosted to ensure offline operation, with each prompt from the EV accompanied by a JSON context file containing data from the TSS server. Responses will be delivered audibly and displayed as a transcript on the UI. For reliability, critical warnings are hardcoded but presented through the AI interface for a unified experience. The AIA continuously monitors telemetry and resource levels from both the headset and PR to deliver predictive safety alerts. Notifications will appear as visual pop-ups with accompanying sound cues and detailed alert text in the corner of the display.

Some vital signals, such as heart rate and oxygen levels, will be monitored continuously by dedicated scripts that trigger alerts when parameters exceed thresholds. The AIA also functions as a task assistant, offering step-by-step guidance for key operations such as LTV search and repair, egress/ingress, and rover navigation. This hybrid framework combines

hardcoded procedures for precision with the LLM's dynamic reasoning to troubleshoot, adapt to unexpected scenarios, and answer astronaut queries in real time.

Artificial Intelligence Assistant (AIA): Chatbot

The AIA is activated via the command "Hey AI." Once activated, the headset listens while the user is talking, then sends audio to our local backend. In our backend, there are 3 steps that must take place before the final audio and text response is sent back to the headset: voice must first be converted to text, then that text must be fed into a Large Language Model (LLM), and finally the text response must again be converted into audio that will be sent back to the headset along with the accompanying text.

To convert voice to text, we will use the "small" version of OpenAI's Whisper which uses 2GB of RAM. If we are low on space or responses must be faster we can use quantized models. For the text to voice, we can either use the Mac's built in Text-To-Speech (TTS) system or the Coqui TTS system. Both do not use significant computational resources. For the LLM itself we plan to use GPT-OSS 20B which uses about 16GB of RAM. The performance of this model is near that of GPT-3 and in our tests, it produced about 500 tokens in 10 seconds on a M5 Macbook Pro. If performance is too slow, we could use Gemma 2 models or Mistral 7B.

The model continuously receives contextual input, including TSS telemetry, system data, location, and prior chat history, allowing it to provide precise, situation-aware responses without relying on external databases or RAG systems. We will fine tune our local LLM to ensure that for direct questions about data, instead of responding with a response like "Your oxygen level is currently at x", the AIA will respond directly with "Oxygen Level: x." Careful prompt engineering will also be employed to maintain consistent, concise responses and minimize redundancy.

Artificial Intelligence Assistant (AIA): Safety

In emergencies such as a sudden drop in life support or off-nominal telemetry, the system shifts to proactively interrupting the astronaut or rover UI with high-visibility warnings and recommended safety actions. Periodically, The AI also assists in time management for procedures based on vitals and navigation paths. Some alerts are triggered by the AI when TSS data is received, while critical warnings remain hardcoded to ensure consistency and avoid hallucinations, maintaining a unified and reliable interface for the EV.

To further reduce hallucinations, we will conduct thorough testing, refine LLM prompts, and evaluate accuracy using a secondary model. If the AI cannot determine a single correct response, it will present multiple options and request human input or prompt rephrasing. EVs will use clear voice commands to engage the AI, with an optional button press available when voice activation is not possible.

Integration with LTV Repair

The AI assistant's primary roles are guiding Exit Recovery Mode procedures, supporting hardware repairs, and performing diagnostics. It will conduct periodic checks on LTV telemetry to forecast potential failures and prioritize high-importance tasks, adapting instructions dynamically based on EV feedback and vitals such as oxygen levels.

By analyzing data from computer vision, TSS, and EV input, the AI will assess the situation and provide optimal or alternative solutions. The LLM converts this information into natural language instructions delivered audibly and visually through AR. During repair, the AI highlights relevant tools and components, answers EV questions, and monitors background telemetry. If a minor issue, such as reduced CO_2 scrubber efficiency, is detected, a yellow

caution alert advises monitoring while confirming sufficient resources to proceed. If conditions worsen, a red warning immediately overrides current tasks, directing the EV back to the habitat and ensuring safety through seamless integration of guidance, monitoring, and navigation.

Integration with Pressurized Rover

The AI Assistant will be integrated into the Pressurized Rover (PR) to support navigation in the DUST environment, guiding search-and-rescue patterns, path planning, object detection, and resource management. A key responsibility of the AIA is to operate within the caution and warning system, monitoring telemetry for off-nominal values and recommending corrective actions to maintain rover safety. To enable efficient and safe search operations for the lost LTV, the AIA will assist in autonomous navigation, calculating optimal paths, avoiding obstacles, and adapting to terrain conditions using real-time data such as LIDAR inputs and updated destinations from the TSS.

Throughout this process, it will provide regular status updates and optimize resource use. After the LTV restart, the AIA will verify updated position data and ensure consistency with TSS readings. Additionally, the AI will assist with PR movement control through natural language commands. The operator can issue verbal instructions such as “move in a circle,” and the AIA will execute the corresponding maneuvers, enhancing usability and reducing manual workload.

E. Human-in-the-loop (HITL) Testing

We will conduct a series of human-in-the-loop tests to iteratively validate system functionality, usability, and reliability across all mission phases. The following tests will be performed:

1. **Display and UI Validation:** Initial testing will verify the ability to project information into the AR space by displaying basic content (“Hello, World”) using Unity. Once confirmed, we will implement a functional UI with static placeholders for core components such as the dashboard and map, followed by dynamic telemetry and real-time data visualization.
2. **Navigation Testing:** The Magic Leap 2 will display maps, plan and trace paths, and allow users to drop pins. Random paths will be generated and tested by two CSI members to evaluate user navigation efficiency under varied lighting conditions.
3. **Egress and Ingress Testing:** Egress procedures will be incrementally tested using the Umbilical Interface Assembly (UIA) checklist displayed through the HMD. Early stages will validate telemetry stream accuracy, while later stages will assess usability as participants follow visual prompts and voice-assisted guidance. Ingress testing will focus on the breadcrumb navigation feature to retrace routes back to the habitat or rover.
4. **Integration with Rover:** These tests confirm synchronization between the rover and spacesuit systems. The Magic Leap 2 must correctly identify and track the rover, ensuring seamless telemetry and procedural data sharing. Testing will be conducted jointly with our partner school’s Rover team using a mock TSS server.
5. **AIA Testing:** AIA tests will be conducted in three phases. Early tests will verify prompt recognition, voice activation, and alert delivery using simulated telemetry data. Mid-phase testing will assess the AIA’s ability to provide context-aware guidance during egress, ingress, and LTV repair, ensuring timely and accurate responses. Final tests will introduce off-nominal scenarios to confirm that hardcoded warnings and recovery

instructions trigger correctly. Participant feedback from CSI members and non-CSI members will evaluate response accuracy and clarity.

6. **LTV Repair Testing:** LTV tests will focus on testing the computer vision capabilities of the software. In particular, we will test whether it is able to detect relevant components on the display. We will also conduct tests to ensure that correct feedback is being relayed to the AIA based on which components have been identified. Lastly, we will ensure that the LTV is able to attain up-to-date data from the TSS regarding those components.
7. **Usability and Unbiased Testing:** Structured usability testing will involve five CSI members unaffiliated with SUITS, completing representative EVA tasks such as navigation, egress, ingress, and basic LTV repair under low-light conditions simulating NASA's Rock Yard environment. To reduce bias, final testing will include 5–10 external participants performing a scripted EVA simulation.
8. **Final NASA Simulation:** The final stage will involve multiple full-system simulations under varied environmental conditions to ensure the Magic Leap 2 and integrated systems perform reliably during the NASA test week.

F. Project Management

Over the course of 5 months, the team will meet for at least an hour every Friday and 2 hours every Saturday with each member being expected to work 1-2 hours outside meetings. At each Friday meeting, the team will discuss work in progress, provide feedback for each other, and assign new tasks to be completed. The team will also be in constant communication with the other asset team and our NASA mentor to ensure that we remain on track. The team will be split up into subteams, such as UI, AI, Nav to gain more expertise and ownership over specific features.

The following plan is from the perspective of a Spacesuit team. Tests refer to those in the HITL Testing section. Appendix B contains a Full Gantt Chart.

Legend (Teams)
All - all team members
UI - user interface
Nav - navigation
Tel - telemetry
AIA - artificial intelligence
LTV - lunar terrain vehicle repair

Month	Who	Planned Action	Planned Outcome
JAN	All	Learn to code in C#, familiarize with Unity and Magic Leap software Coordinate with the PR team and NASA mentor	Code in C# proficiently Plan goals and meetings
FEB	UI Nav Tel	Design rough draft of UI elements: UIA procedures for EVA, 2D map, warnings, EV suit and biomedical data, navigation paths, and points of interest. Code a skeleton program outlining the major algorithms to be written later, including: best path navigation, pin drops on the map, and breadcrumb path. Ensure data can be transferred between assets. Test TSS' update frequency and accuracy. Identify data needed for UI elements.	Pass "Display and UI Validation". A rough code framework addressing mission aims is established. Data can be transferred between components

	AIA	Establish the AIA's core framework and initial functionality. Finalize and implement selection of model architecture. Implement MCP tool interface in Unity. Integrated placeholder text and voice outputs. Create mock AIA.	Functioning mock AIA can answer 2-3 preset procedural queries
	LTV	Develop the foundation for the LTV Repair system. Collect and preprocess a dataset of damaged and intact LTV components. Train and evaluate a DINOv3 backbone with a linear probe for binary classification. Integrate the model with Unity to display red fixed-point overlays. Coordinate with the AIA team to align ERM and repair tasks.	Model demo that classifies sample LTV images and outputs bounding overlays
MAR	All	Write code to achieve mission aims (as outlined by the Requirements section of Mission Description). Begin testing in conditions close to that of the described EVA scenario.	Each algorithm is completed and accomplishes mission aims
	Nav	Navigation algorithms written and compatible with UI elements.	Pass “Navigation Testing”
	Tel	Collaborate with PR team to ensure telemetry data and other shared elements are planned	Pass “Ingress and Egress”
	AIA	Transition from mock AIA to a functional prototype operating on simulated TSS data	Pass “AIA Testing”
	LTV	Implement live, frame-by-frame damage detection integrated with AIA-guided repair instructions, producing an interactive repair demo	Pass “LTV Repair Testing”
	UI	Implement designs and test interactability of designs in our Unity project. Including: minimap, navigation details, palm-activated menu, biomedical data, and warnings.	Necessary UI components are functional in variable lighting conditions
APR	All	Test with PR team and external evaluation	Revise product based on third-party feedback
		Final adjustments and polishing.Tweak product design based on evaluation feedback. Fix hiccups/bugs discovered during testing. Refine product appearance.	Pass “Usability and Unbiased Testing” and “Integration with Rover”
MAY	All	Have everything polished a week before test week: - Review code - Deploy and submit product	Ensure everything is ready to go. Pass “Final NASA Simulation”

G. Technical References

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II. Outreach Plan

Columbia's NASA SUITS Team operates within the Columbia Space Initiative (CSI), a multidisciplinary, student-run organization dedicated to advancing space science and engineering. As part of CSI, the team contributes to one of the largest and most consistent STEM outreach programs in New York City, engaging over 1,000 K–12 students each year across the greater New York area. CSI's outreach efforts focus on underserved and underrepresented communities in STEM, bringing hands-on learning experiences to students. Through these programs, Columbia students aim to inspire curiosity, promote equitable access to STEM education, and prepare young learners to become future contributors to space exploration.
This year, Columbia's SUITS team will take a leading role in expanding CSI's outreach curriculum, placing a particular emphasis on computer science education with a human-computer interaction and UI/UX emphasis. In addition to supporting CSI's established outreach pipeline, the team will design new modules that share the work and outcomes of the NASA SUITS Challenge directly with students. These lessons will highlight how emerging technologies, such as augmented reality, AI assistance, and intuitive user interfaces, enhance astronaut performance and mission safety.
Columbia's SUITS team not only extends NASA's mission of inspiring the Artemis Generation but also strengthens the bridge between academic research, technological innovation, and community education ensuring that the excitement of human spaceflight reaches students from all backgrounds.
CSI's Outreach team has many partnerships with schools, non-profits, and other educational organizations across New York and the US. Some of the team's partners this year include:

- Sophie Gerson Healthy Youth (SGHY), a NYC non-profit dedicated to supporting underserved middle schoolers at 5 schools in the city
    - Within the SGHY umbrella is the immersion cohort, a small group of students at each SGHY school who take part in more intensive space science programming
- The Columbia Engineering Outreach Office, an internal organization that facilitates educational outreach events with local high schoolers
- Eagle Academy for Young Men of Harlem, an all-boys middle and high school in East Harlem

The team's efforts span several programs, including the new SUITS curricula and lessons. Each of these programs is described briefly below.

SUITS Coding Curriculum

The SUITS Coding Curriculum teaches students the importance and functions of coding, as well as the basics of coding through an interactive activity: using Scratch to build moon-themed projects and games. In a weekend event for middle school students, SUITS team members will provide a tutorial on game development in Scratch, with an emphasis in UI and interactive software. The in-person 2026 spring event will aim to inspire interest in programming and space-focused technology.

Spaceposium

At the end of the academic year, Outreach, along with the greater CSI community, holds Spaceposium, a culminating event to highlight the year's efforts and successes. Middle schoolers

from our partner schools are invited to engage in poster sessions held by various missions of CSI, including SUITS. This past spring, Outreach also conducted activities geared toward the students such as the Physics Plasma Lab tours, an activity about biomedical engineering, and the aforementioned robotics activity.



SUITS 2025 members presenting at Columbia’s Engineering Showcase (left) and Spaceposium (right)

Auditorium Programs

Through the team's partnership with SGHY and Eagle Academy, CSI has visited six middle and high schools in the area several times over the past few years, and is of course continuing this year. Each visit consists of a space-focused STEM lesson, complete with an interactive activity to allow students to directly engage with the topic. Throughout each educational lesson and activity, CSI emphasizes the hands-on nature of STEM, and works to convey every club member's passion for space exploration. These programs take multiple forms, highlighted in the below three subsections.

Circuits Activity

The Circuits Activity auditorium program, which was debuted by CSI this past summer at the annual SGHY Summer Camp and is especially related to SUITS, consists of a lesson on basic circuits and their uses, as well as an activity (the NASA Chandra Light Up Exploded Stars and Constellations activity). During this year, SUITS members will help guide the Circuits Activity and teach students how circuits make SUITS possible.

Aerodynamics Auditorium Program

The most popular of these auditorium programs is a hands-on lesson on the principles of aerodynamics. Following a short lecture on fundamental concepts, students apply their aerodynamics knowledge by designing, building, and launching paper airplanes in an attempt to hit small drones and compete for the farthest launch.

Water Balloon Drop

The Water Balloon Drop auditorium program is a structural engineering challenge. After learning about crumple zones and how parachutes work, students are prompted to design and build a structure to prevent their group's water balloon from bursting from a 10-foot drop. We challenge students to use as little material as possible, considering the real-life engineering challenge of working with limited resources.



Photos of various auditorium program activities from previous academic years

Model Rocketry

Outside of these recurring visits, CSI will also continue its model rocketry program this upcoming year. The program includes a digital curriculum that consists of volunteer-created video lessons and interactive activities. These lessons provide instruction on aerodynamics, propulsion, and the impetus behind space exploration (including information about the Artemis program). The program culminates with a model rocket launch where students showcase what they've learned with custom-designed and decorated rockets. Thanks to further funding from the New York Space Grant, these same programs are also implemented at Eagle Academy in addition to our SGHY partner schools.



Photos from a model rocketry launch at 75 Morton

Science Saturday at Rockefeller University

CSI participated in Rockefeller University's annual Science Saturday! Here, CSI members presented to about 200 elementary and middle schoolers from around NYC modified versions of three of our programs — the Circuits Program, Model Rocketry Program, and Aerodynamics Auditorium Program.




Photos of CSI Outreach at Science Saturday! Photos courtesy of Matthew Septimus.

CubeSat Curriculum

Additionally, as a part of NASA's CubeSat Launch Initiative grant (which was recently won by CSI's Cube Satellite team), SUITS team members will contribute to the implementation of a brand new CubeSat outreach effort aimed at educating middle and high school students about satellites. The program, to be debuted this spring, will provide access to hands-on examples and activities to build STEM knowledge and excitement in students underrepresented in STEM. This multi-year course will also include open-access digital lessons, and in-person visits to schools.

Sophie Gerson Healthy Youth Summer Camp

CSI members serve as Space Science Counselors during SGHY's yearly week-long summer camp, which is conducted every August. During the summer camp, CSI conducts daily auditorium programs, rocket launches (combustion and water rockets), nighttime astronomy sessions where students get to work with telescopes, and debuts a new curriculum.






Photos from August 2025 Camp.

Other Events

Below, you will find further information on a number of formally planned outreach events with SGHY and Eagle Academy in the next year, as well as smaller, one-time supplemental events that CSI will also be hosting. Further down are correspondences with Alan Gerson, the director of the Sophie Gerson for Healthy Youth, and Andrew Peterson, our contact at Eagle Academy, as well as photos from previous outreach events.


A collection of photos from previous years' Spaceposium events. Bottom right: Middle school students presenting to professor and former NASA astronaut Mike Massimino.

2025- 2026 Events Timeline

Event	Number of events	Notes/Description	Dates

Education First Tours	~4 expected	CSI volunteers give engineering tours for middle and high school students to help fund other programming	January–August 2024 (none scheduled)
Auditorium Programs	6	Space-themed STEM lessons and activities (Aerodynamics and Circuits)	Fall 2025 and Early Spring 2026
SUITS Coding	1	SUITS Specific Outreach	Spring 2026
Rocketry Launches	6	Visit to facilitate launches at each school	Early to Mid Spring 2026
CubeSat	4–6	First auditorium programs with schools	Spring 2026
Spaceposium	1	End of year CSI event presenting our projects to middle school students, running engineering activities, hosting a Q&A with an astronaut (club advisor), and touring the engineering labs	April 2026
SGHY Summer Camp	1	CSI volunteers work at SGHY-funded summer camp as astronomy counselors	1 Week in August 2026

Below are correspondences from Alan Gerson, President of Sophie Gerson Healthy Youth confirming their intent to work with us this year. All emails are with Ren Kirchmann and Maryam Naser, CSI's broader Outreach leads.

---------- Forwarded message ---------
From: **alan jay gerson** <alangerson@gmail.com>
Date: Mon, Oct 27, 2025 at 3:57 PM
Subject: Here you go...
To: Ren Kirchmann <ren.kirchmann@columbia.edu>, Maryam Naser <mn3171@columbia.edu>

This Message Is From an External Sender
This message came from outside your organization.

Hi Maryam, Ren,

Once again, all five Sophie Gerson Healthy Youth partner schools have requested the CSI program! This reflects CSI's popularity among students and recognition by our educators of the value you bring to their students! You provide role models for our students to emulate. CSI's hands-on activities inspire greater interest among our students in science generally and space science particularly, and the self-confidence that they can do science and grow up to become scientists. Your presentations also imbue an awe at the majesty of the cosmos and the possibilities of space exploration.

As you know, all Sophie Gerson schools consist of middle schools, grades 6, 7, & 8 (roughly ages 11 - 13 - perfect ages for inculcating an interest in space science). All five schools include children from very low income households, including children living in homeless shelters. Their parents lack the means to take them to museums or planetariums or provide science kits. Those students would lack the above inspiration without CSI coming to their schools and welcoming them on campus. Three of our schools consist of Title One schools, in which virtually all students live in households at or below the poverty line. Our other two schools, more economically diverse, include such children. Once again, here are the schools and school contacts:

MS 302, 681 Kelly Street, Southeast Bronx, Rosny Louis, Assistant Principal, 516-430 -0339;

MS 126, 80 Catherine Street, Chinatown/Lower East Side, John DeMatteo, SGHY Lead Teacher, 917-826-2269;

MS 140, 123 Ridge Street, Lower East Side, Channell Jenkins, Assistant Principal, 917-312-8835;

MS 167, 220 East 76th Street, Upper East Side, Charissa Hinnant, Assistant Principal, 732-766-9554;

MS 297, 75 Morton Street, Greenwich Village, Lauren Brooks, SGHY Lead Teacher, 914-525-1952.

Our schools and we feel especially excited that this year's program will center around the April 1 launch of your CubeSat, with our middle school students serving as citizen scientists. As discussed, CSI will provide each school with two auditorium programs prior to April 1. Those programs will include hands-on activities which introduce students to CubeSat and its launch, including the principles of air and space flight, circuitry, life on the International Space Station which receives your CubeSat, and problem solving relating to the design and composition of CubeSat. On or shortly after April 1 we will livestream your CubeSat launch or ,depending on the launch time, provide video feed of it, with CSI commentary. After May 1, when the CubeSat starts transmitting data, CSI will provide another session in which students receive and analyze the data, followed by the program's grand finale: launching of model rockets the students prepared. In between, CSI members will accompany students on their school field trips to the Intrepid Museum and the Planetarium, and welcome a smaller group selected by the schools to the CSI Spaceposium on campus.

Each auditorium program will run for 45 -60 - 90 minutes, depending on particular school logistics. An entire grade, 80 - 120 students depending on the school, will attend. At the Wagner school, on E. 76th Street, you will conduct two back-to-back auditorium programs, due to the school's student population size, with around 120 students in each. Classroom teachers will follow-up with in-class lessons based on your presentations.

We look forward to "launching" the program! I will get back to you later this week with proposed schedules.

Thanks for your interest in and commitment to spreading the joy and value of space science and science in general.

Best,

Alan

Alan J. Gerson, President
Sophie Gerson Healthy Youth
sophiegersonhealthyyouth.org
917-836-3272

III. Administrative Section

Institutional Letter of Endorsement

COLUMBIA | ENGINEERING
The Fu Foundation School of Engineering and Applied Science

October 28, 2025

Proposal Review Committee
SUITS 2026 Design Challenges
Johnson Space Flight Center
Houston, Texas 77058

Re: NASA SUITS Design Challenges 2026

Dear Members of the Review Committee,

The Columbia Space Initiative is a student-led organization within Columbia University in the City of New York. Its faculty sponsor is Michael Massimino, a Professor in the Department of Mechanical Engineering.

The Fu Foundation School of Engineering and Applied Science endorses the team's participation and commits to supporting the participation of the Columbia Space Initiative in their Challenge.

Please do not hesitate to contact me or Professor Massimino should you have any questions or concerns.

Sincerely,

Hod Lipson

Hod Lipson
James and Sally Scapa Professor of Innovation in the Department of Mechanical Engineering
Co-Director of Columbia Makerspace
Chair, Department of Mechanical Engineering

Statement of Supervising Faculty

COLUMBIA | ENGINEERING
The Fu Foundation School of Engineering and Applied Science

October 28, 2025

Proposal Review Committee
SUITS 2026 Design Challenges
Johnson Space Flight Center
Houston, Texas 77058

Re: NASA SUITS Design Challenges 2026

Dear Members of the Review Committee,

As the faculty advisor for an experiment entitled "Lunar Lions" proposed by a team of undergraduate students from Columbia University, I concur with the concepts and methods by which this project will be conducted. I will ensure that all reports and deadlines are completed by the student team members in a timely manner. I understand that any default by this team concerning any program requirements (including submission of final report materials) could adversely affect selection opportunities of future teams from their institution.

Sincerely,

Michael J. Massimino

Michael J. Massimino
Professor of Professional Practice

The Fu Foundation School of Engineering and Applied Sciences

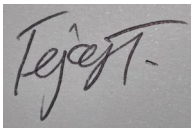
Statement of Rights of Use

As a team member for a proposal entitled “Lunar Lions” proposed by a team of higher education students from Columbia University, I will and hereby do grant the U.S. Government a royalty-free, nonexclusive and irrevocable license to use, reproduce, distribute (including distribution by transmission) to the public, perform publicly, prepare derivative works, and display publicly, any technical data contained in this proposal in whole or in part and in any manner for federal purposes and to have or permit others to do so for federal purposes only. Further, with respect to all computer software designated by NASA to be released as open source which is first produced or delivered under this proposal and subsequent collaboration, if selected, shall be delivered with unlimited and unrestricted rights so as to permit further distribution as open source. For purposes of defining the rights in such computer software, “computer software” shall include source codes, object codes, executables, ancillary files, and any and all documentation related to any computer program or similar set of instructions delivered in association with this collaboration.

As a team member for a proposal entitled “Lunar Lions” proposed by a team of higher education students from Columbia University, I will and hereby do grant the U.S. Government a nonexclusive, nontransferable, irrevocable, paid-up license to practice or have practiced for or on behalf of the United States Government any invention described or made part of this proposal throughout the world.

Michael Massimino

Michael Massimino

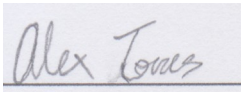


Tejas Tirthapura

Hiba Altaf

Grace Xu

Andrew Bonilla



Alex Torres

Nicole Cui

Elizabeth Liu

Thea Mulia

Priya Sinha

Emily Ta

Ava Kim Cohen

Grace Cho

Mia Ravishankar

Alexander Tu

Lin Pan

Emma Krgo

Richard Li

Precious Jepkorir

Yumeng Zou

Funding and Budget Statement

Item	**Projected Cost**
Item	**Projected Cost**
Materials and Supplies	
Magic Leap 2	$4000
Development Software Package costs (OpenCV, Text to Speech, etc)	$500
Other	$200
Materials and Supplies Total	$4700
Travel	
Airfare for 8	$4,800
Hotel (4 Rooms 4 Nights)	$3,500
Car Rental	$300
Food ($50 per person per day)	$2,000

Hololens2 Loan Program

We have a device but would still like to be considered for a loan to aid in our development.

Appendix:

Appendix A

Member Emails

Grace Xu	gx2168@columbia.edu
Hiba Altaf	ha2616@columbia.edu
Andrew Bonilla	acb2319@columbia.edu
Richard Li	rjl2194@columbia.edu
Lin Pan	lp3166@columbia.edu
Ava Kim Cohen	akc2214@columbia.edu
Mia Ravishankar	msr2230@columbia.edu
Christopher Cao	cc5450@columbia.edu
Sreya Chakraborty	sc5382@columbia.edu
Emma Krgo	eak2235@columbia.edu
Alex Torres	adt2154@columbia.edu
Hudson Miller	hm3086@columbia.edu
Adriano Rangel-Guzman	ar4918@columbia.edu
Henry He	sh4421@columbia.edu
Thea Mulia	tm3468@barnard.edu
Precious Jepkorir	pmj2129@columbia.edu
Emanuel Antunez	ema2212@columbia.edu
Yuhang Qiu	yq2438@columbia.edu
Alexander Tu	awt2138@columbia.edu
Alice Lin	al4937@columbia.edu
Claire Yang	qy2313@columbia.edu
Elizabeth Liu	el3206@columbia.edu
Priya Sinha	ps3491@barnard.edu
Yilin Julia Wang	yjw2109@columbia.edu

Appendix B

Gantt Chart

Project Name: CSI SUITS 25/26
Project Start: 1/1/2026
Current Date: 1/1/2026
Weeks In Progress: 0

Task	Subteam	Start Date	End Date	Days	Progress
Planning Stage					
Learn how to code in C#	All	1/1/2026	1/31/2026	30	0%
Familiarize with Unity and Magic Leap software	All	1/1/2026	1/31/2026	30	0%
Begin coordination w/ the PR team and NASA mentor	All	1/1/2026	1/19/2026	18	0%
Execution Stage					0%
Outline and design rough draft of UI elements	UI	2/1/2026	2/28/2026	27	0%
Code skeleton program outlining major algorithms	Nav	2/1/2026	2/28/2026	27	0%
Ensure data can be transferred between assets	Tel	2/1/2026	2/28/2026	27	0%
Establish AIA core framework and initial functionality	AIA	2/1/2026	2/28/2026	27	0%
Develop LTV Repair system foundation	LTV	2/1/2026	2/28/2026	27	0%
Write code to achieve mission aims	All	3/1/2026	3/31/2026	30	0%
Collaborate with PR team to ensure telemetry data an	Tel/All	3/1/2026	3/31/2026	30	0%
Functional AIA prototype	AIA	3/1/2026	3/31/2026	30	0%
Interactive repair demo	LTV	3/1/2026	3/31/2026	30	0%
Begin testing in conditions close to EVA scenario	All	3/1/2026	3/31/2026	30	0%
Review Stage					0%
Test w/ PR team and external evaluation	All	4/1/2026	4/30/2026	29	0%
Adjustments and polishing	All	4/1/2026	5/15/2026	44	0%
Final Polish	All	5/15/2026	5/22/2026	7	0%



Appendix C

Additional Images

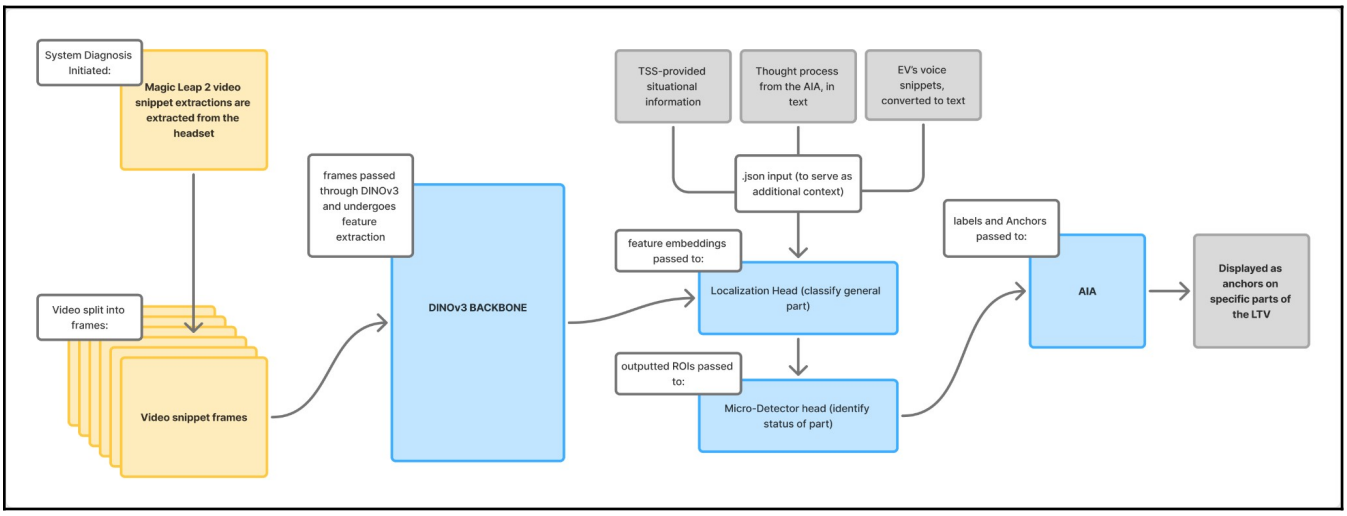


Fig. 7 LTV Architecture

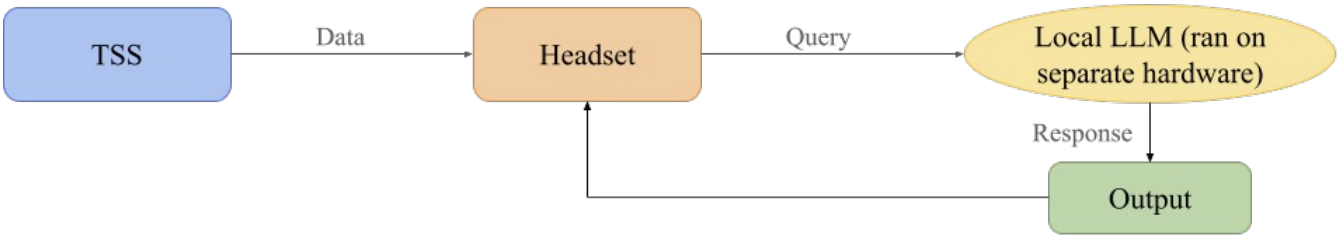


Fig. 9 System Architecture

Concept of operations (original)

C. Concept of Operations (CONOPS)

Egress

Our mission's objective is to ensure a safe and efficient astronaut egress from the habitat, beginning within the airlock. Here, the crew member dons the HMD, spacesuit, and other EVA tools before initiating the Egress Mode. Once activated, the HMD transitions into Egress Mode, guiding the astronaut through all egress procedures with an emphasis on intuitive user interaction.

A collapsible, interactive task list that presents the Umbilical Interface Assembly (UIA) Egress procedures in a clear, step-by-step format. Tasks dynamically expand or collapse to minimize visual clutter and allow the astronaut to focus on their current operation. Each step in the checklist corresponds to telemetry feedback from the UIA. When a switch or setting is confirmed through the Telemetry Stream Server, the interface provides audio and visual checkbox confirmation. If a step is completed out of order, skipped, or performed incorrectly, the system immediately issues a notification warning, alerting the astronaut both visually and through a brief voice cue to ensure procedural integrity.

In addition to the procedure checklist, an audio command is given for each current procedure task and an overlaid UI is placed on top of the UIA to signify the location of the switch to reduce cognitive load for the astronaut. The astronaut can interact with the interface hands-free using audio commands, allowing them to progress through steps, request status updates, or expand checklist sections without manual input. If clarification or troubleshooting is needed, the astronaut can ask the AI assistant directly. The AI system is capable of explaining UIA terminology and providing real-time procedure guidance.

EV Navigation

After completing the egress procedures checklist, the HMD of the EV whose primary role is navigation will transition to its default state. This default state prioritizes navigation features and presents a customizable user interface with various widgets and software, designed to enhance visibility in the challenging low-light conditions of the lunar south pole.

We plan to employ AIA in EV Navigation by using the LLM to assist our path-finding algorithms in determining optimal paths. The AIA can provide the user with verbal and written instructions on how to best follow the generated path(s) and potentially offer the user multiple navigation paths to select from. In addition, we can use computer vision to have the LLM analyze the surroundings and draw the generated path overlaid on the screen. Given the space suit life support and limitations, the AIA can predict the maximum range of usage, which will help determine our generated navigation path. As a part of the AIA safety alert system, the AI will notify the EV when biometric telemetry is off nominal

ranges via visual cues and auditory warnings. The AI system will then guide the astronaut on how to get back to acceptable ranges. The AI will provide similar warnings when the EV encounters dangerous terrains and hazards.

This navigation state features an AR overlay, providing real-time situational awareness by displaying essential data directly in the astronaut's field of vision. This includes obstacle detection alerts, waypoint guidance, and a toggleable path feature. A breadcrumb feature allows astronauts to retrace their steps back to their starting position. Overlapping paths will be removed to reduce visual clutter and simplify navigation. In addition, an active mini-map widget shows the user's location relative to surroundings, the location of hazards, and updates in real-time to offer a comprehensive view of the operational area. When the map is interacted with, it will become enlarged on the screen, displaying more details such as more of the surrounding area, sites of importance, etc. In this enlarged mode, the user can drop a pin of their current location or a location of interest to be viewed on both the mini-map and its enlarged version.

In low-light conditions, thermal infrared cameras power a heat map that highlights nearby terrain features and obstructions. This is complemented by advanced computer vision capabilities that enable real-time recognition of hazards such as rocks, craters, and irregular terrain. The system provides autonomous path planning, continuously analyzing the terrain and updating the recommended route based on real-time obstacle detection. The software system also implements comprehensive tracking of crew and scientific assets in the field. It provides real-time updates on asset locations and status, accessible through the HMD interface.

In case of emergencies or the sudden appearance of obstacles, the system automatically recalculates the best alternative route. These updates appear on the crew members' HMD, allowing for quick adaptation without manual intervention. The system also provides customizable alerts for various scenarios, including proximity to hazards, deviation from planned routes, or updates on asset status.

LTV Repair

The EV arrives at the malfunctioning LTV and verbally commands AIA to begin a system diagnosis, utilizing the ERM procedures, TSS information, and DINOv3 model (pipeline described in Hardware/Software). The outputted DINOv3 anchors will be displayed thru to the frontend, from which the AIA will guide the EV to conduct visual inspection and repair tasks. After EV commands AIA to begin system repair, AIA first begins guiding EV through procedures of physically restarting the rover navigation system. TSS will indicate to AIA when restart is complete, during which AIA will check if reset local rover position data matches TSS position data. If not, AIA will warn EV of the position mismatch and ask EV to set the rover position to the last TSS position. The system diagnosis process iteratively repeats until all warnings are addressed, with each previous diagnosis being used as feedback to improve the next diagnosis. Upon completion of the repair process, AIA will instruct EV to conduct a final system diagnosis using the same concurrent visual and model inspection process to

ensure stable recovery of the rover. EV then verbally confirms system diagnosis completion after all system checks.

Ingress

After completing the LTV Repair, the user will enable Ingress Mode. In Ingress Mode, the astronaut is directed back to the habitat using the shortest and safest route, displayed in both the augmented reality breadcrumb path and on a mini-map within the HMD. The route continuously updates as new obstacles are detected, ensuring the astronaut maintains safe navigation back to the airlock.

When the astronaut reaches the habitat, an Ingress Procedures checklist automatically activates. A collapsible, interactive task list like the one for Egress Procedures appears in the HMD's peripheral view, minimizing obstruction while guiding the astronaut through the re-entry sequence, and each step corresponds to telemetry data from the Telemetry Stream Server (TSS). When a switch or action is successfully completed, the interface provides audio and visual checkbox confirmation.

Just like during Egress, incorrect step warnings, audio commands, overlaid UIA highlights, and AI voice assistance is used to properly guide crew members through the Ingress tasks. Once all checklist objectives are complete, and all systems are confirmed nominal, the EV may operate the airlock and safely re-enter the habitat.

Pressurized Rover Navigation

The Rover will be instructed by the Lunar Terrain Vehicle (LTV) to an area of geologic interest. The rover will then calculate the safest, shortest route to the destination, while simultaneously updating as hazards and obstacles appear. The rover will also estimate turnaround points based on the amount of remaining resources. This will ensure that the astronauts can safely return to the habitat.

The rover has a life support system that tracks the EV's biostatistics, which are displayed on the main interface. If the vitals are not ideal, a warning will pop up, notifying the user to return to the habitat. The rover interface will also constantly display battery level to determine if the mission can be successfully completed. If the battery level is not sufficient, the rover will be instructed to go back.

Continuous Tasks

The EV will be able to attach current tasks and procedures to the main screen via the hand menu. This is also where the EV can open up their biometrics and suit data. Data about oxygen levels, CO_2 levels, pressure, temperature, and humidity inside the suit will be available. If desired, the mission timer will also be available in the hand menu, reducing the number of icons on screen at all times. The timer will display messages at key points in the mission, letting the EV know how much time they have left to complete a specific task to stay on track.

Original navigation

Navigation Algorithm

A major impediment to navigation in this mission is the fact that the Pressurized Rover (PR) begins with incomplete knowledge of its environment. As such, our aim is to implement an algorithm that allows the rover to gather real-time information about its surroundings and procedurally construct an optimal path towards its goal as it acquires more knowledge. The D* Lite algorithm is well-suited for this task. D* Lite performs a backwards search from the goal to the agent and starts by assuming that the environment is obstacle free. As the agent learns about its environment, the algorithm propagates the costs of any obstacles back to its start node and allows the agent to iteratively reconstruct its path in realtime. Importantly, it guarantees the optimal path while being computationally fast and memory efficient [1].

We will map out the region using a weighted graph with nodes and edges marking locations on the map and the cost to reach said locations. The cost function *g(n)* will take into account spatial distance, resource usage, and slope to determine the total cost from node *n* to the goal. We will also calculate a one-step lookahead function *rhs(n)*, which estimates the minimum cost to the goal via *n*'s neighbors. The algorithm uses *rhs(n)* and *g(n)* to eliminate inconsistent nodes and revise the optimal path. For our heuristic function *h(n)*, we will use the Euclidean distance from node *n* to the start node, and calculate the travel cost using the minimum weight, assuming a perfectly flat and uniform environment. The Euclidean distance applies to this situation since the rover is omnidirectional and can move freely within the space. This heuristic will always underestimate the actual travel cost and is, therefore, admissible.

If the rover detects a hazard, it will give the location an exorbitantly large cost, and the rover will naturally avoid it when determining the optimal path. The particular method of identifying hazards is discussed in greater detail later in our proposal.

Search and Rescue

Our rescue protocol will begin by having the PR navigate towards the last known location of the LTV using D* Lite as described above. Once it reaches this point, it will commence the next phase of the search and rescue by performing an expanding square search within the radius defined by the LTV's consumable limits. The radius can be calculated using the LTV's max speed and battery information, and will reduce the required search space. To ensure we do not miss the proximity signal from the LTV, the PR will move in units of 950m, which is below the signal diameter of 1000m, as stated in the proposal. At the location the signal is detected, which we will call point A, the PR will begin doing a more refined expanding square search centered around A. The PR will cover a radius of 500m around A and move in units of 10m, taking advantage of the LIDAR and beacon signals to locate the LTV.

The PR will use D* Lite to navigate to the designated points in the search pattern to ensure that it properly evades obstacles and accounts for the terrain.

When the PR comes within 50m of the LTV, an indicator will appear on the screen showing the proximity between the two objects. The indicator will relay its GPS coordinates, which will allow the system to navigate the rover towards the LTV. The icon will be accompanied by a beeping sound that becomes more frequent as the PR approaches the LTV, somewhat like a metal detector.`

Hazard Identification and Avoidance

To pass environment information to the D* Lite algorithm, we will employ computer vision techniques, which will process images and live video from the rover's cameras to map out the space, and detect and classify hazards along the intended path (shown in blue in Fig. 1). The TSS makes the rover video stream easily accessible, along with GPS information like its x and y coordinates, incline, and direction, which we can use towards constructing our map. In addition to rover cameras, live video streams from crew members' cameras can assist in building other portions of the map.

Once a rover has identified a hazard, our algorithm will determine the object's dimensions and place it on our map, with the hazard being denoted by an extremely large cost to reach that respective node on the map. We will then recalculate the path using D* Lite so that it avoids the hazard.

Resource Tracking

Before the PR can set out on a particular journey, it must determine whether the journey is possible given its power level and other vital information like oxygen levels. If the cost for reaching the goal exceeds the maximum possible traversal distance and time given the parameters, the path should be aborted, and the PRSU will be informed that such a journey is impossible unless certain battery/vital levels are met. To determine how quickly resources will deplete, we will consider factors like the distance.

To prevent resource depletion and estimate turnaround points, we will model the terrain as a graph with nodes representing potential turning points and edges representing the costs (resources consumed to navigate to those nodes). Using D* search on this graph with an initial maximum cost constraint, we can predict turning points to ensure the rover returns before exceeding resource limits [2]. In addition, hazards detected by both the rover's video stream and the crew members' vision are automatically updated onto the shared map, allowing future navigation tasks to leverage these discovered hazards for safer and more efficient path planning.

The rover's battery level will be tracked by an algorithm that periodically assesses whether the current power is sufficient to complete the journey. For long-term resource prediction, the algorithm considers factors like the distance to the goal, terrain slope (learned over time by tracking the rover's angle, heading, and speed), and time of day (if the rover is solar-powered). The algorithm updates predictions as the optimal path changes.

Breadcrumb Path

A breadcrumb feature will enable astronauts to visually retrace their steps back to the PR, enhancing safety during exploration. The system automatically places breadcrumb markers at regular spatial intervals along the astronaut's traveled route, forming a continuous path overlay anchored to the lunar terrain.

Overlapping segments are dynamically consolidated to minimize visual clutter, maintaining spatial information without overwhelming the user. Each new marker is checked against the last placed marker using a predefined distance threshold (e.g., 1–2 m). If the new marker falls within this proximity, it is either skipped or merged with the previous marker, ensuring that overlapping markers do not accumulate. Each breadcrumb carries a timestamp, enabling the rendering system to gradually fade opacity or shift color over time to indicate recency. Shader-based rendering pipelines handle these effects efficiently, interpolating alpha and color values in real time. An overhead compass with waypoints will be included to highlight the direction that the astronaut is heading and to allow them to easily orient themselves on the map.

The breadcrumb system will integrate with the broader navigation framework that computes optimal travel paths based on environmental and physiological data. This integration allows the breadcrumb to not only record and visualize previously traveled routes, but also to reference the computed best-path information when guiding the astronaut back toward the PR. For instance, if the navigation system determines a safer or more efficient return path, the breadcrumb overlay can adapt accordingly, highlighting alternative segments, fading obsolete trails, or suggesting minor course corrections.